

Special Types of Memories

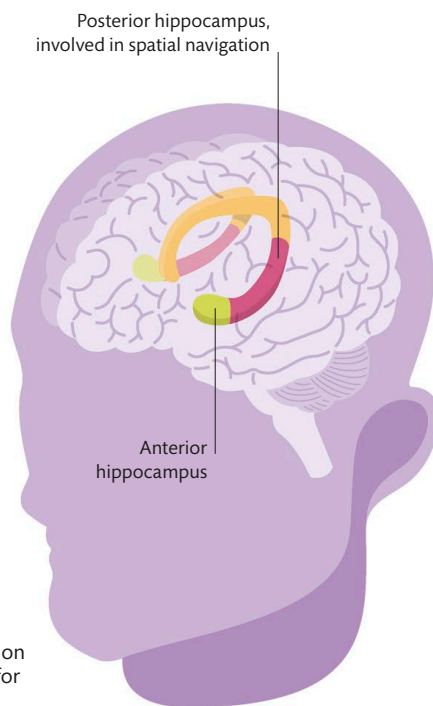
Although a few children exhibit remarkable skills, most people with exceptional memory are not born that way. Instead, they use special techniques and lots of practice, sometimes leading to physical changes in their brains.

Training exceptional memories

Scientists studying trainee London taxi drivers as they learned “the Knowledge” (a huge network of roads and landmarks) found that the volume of the subjects’ posterior hippocampi increased as their ability to navigate improved. This could occur due to the birth of new neurons or the growth of existing dendrites (see p.20). However, the taxi drivers performed worse than control subjects in memory tests not involving London landmarks. This suggests memory is finite, and improving one area may come at the expense of others.

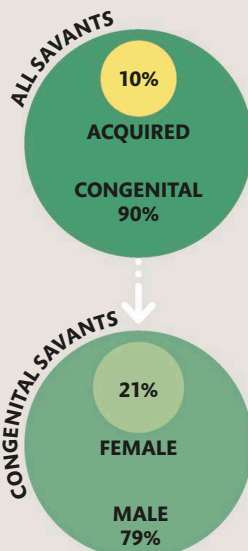
Hippocampal structures

Our two hippocampi—one on each side of the brain—are vital for learning and memory. They can be divided into posterior (back) and anterior (front), with the posterior portion particularly important for spatial navigation.



Savant syndrome

People with mental disabilities sometimes demonstrate incredible abilities in one specific area, often related to memory. This is called savant syndrome. Many savants are autistic, but the syndrome can also be triggered by severe head trauma. Some savants can calculate the day of the week for any given date. Others remember everything they read or can paint detailed pictures of scenes they have seen only once. Scientists think these talents may develop because of savants’ extreme focus and interest in one area. There is also evidence they see the world as building blocks, not whole pictures, by accessing perceptual information most of us are not consciously aware of.



By genetics and gender

One database of savants, as reported by their parents or caregivers, found that the vast majority (90 percent) are born with the condition, and of these, most were male.

FLASHBULB MEMORIES

People often remember where they were when receiving emotional news, and the memory seems extremely vivid and detailed. These are called flashbulb memories. However, studies have shown that we are as likely to be mistaken about these snapshots as we are about any other memories.

